JM0211 Process Mining

Conformance Checking for Process Improvement

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Motivation and Scientific Background

- **Use case.** Dutch bank, BPI 2017: 3,093 denials, under 1 day to ~260 days
- **Question.** Which deviations go with slow denials? Allow or forbid?
- **Not all bad.** Some are workarounds (Dees et al., 2017)
- **Course fit.** We check **control flow** (order of activities) against a **process model**; other course topics check resources or time

3,093	54	14.43	fast 1,501 / slow 1,592
denied applications	visible activities	days, median throughput	cutoff 14 days

Explanation of Conceptual Approach

Input XES log + Petri net (PNML) › **1 KPI** Throughput time › **2 Align, encode** Count log/model moves › **3 Classify** Slow > 14 days; tree › **Output** Rules (one per tree leaf) + trace lists (CSV)

54 activities x 2 move types = 108 features

Hold-out 25%, seed 0. Grid: 48 combinations. 5-fold CV on F1 of the slow class

Step 2: trace A, A, C vs model A, B, C

	1	2	3	4
Trace	A	**A**	**>>**	C
Model	A	**>>**	**B**	C
Move	sync	**log move A**	**model move B**	sync

log move = extra step in the case,

model move = step the model expects but the case skipped,

>> = no step

Evaluation Design and Results

1. Correctness — 2 hand-built toy logs (5 and 6 traces) › **our pipeline** › All alignments exact (11 of 11 traces). Both planted rules recovered. Precision = recall = 1.0

2. Usefulness (slow class) — real log: 3,093 traces, 25% hold-out › **our pipeline** ›

	Precision	Recall	F1	Rules
All-slow baseline	0.515	1.000	0.680	1
Tree, seed 0	**0.563**	**0.970**	**0.712**	**2**
Tree, mean seeds 0 to 3	0.582	0.855	0.688	2 to 14

Main finding: seed 0 tree (base rate of slow 51.5%)

Was the call after the offer paused more often than the process allows?

No: 389 traces, 16% slow → **fast**

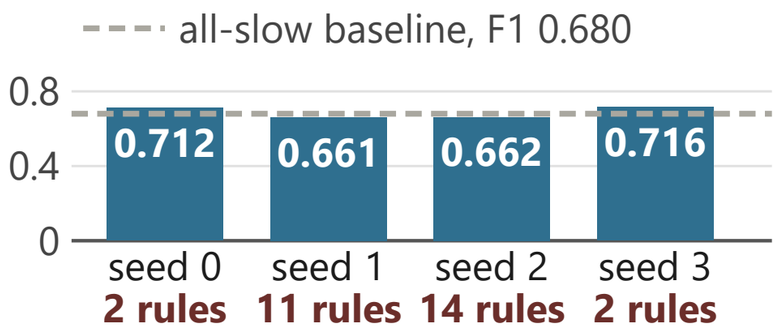
Yes: 2,704 traces, 57% slow → **slow**

`rule feature: log move of W-Call-after-offers-suspend`

Call after offers = the bank's follow-up call after sending an offer. Paused = the work item is put on hold.

One pause is allowed. With an extra pause 57% of cases are slow, without it 16%.

14 days: four seeds, F1 (mean 0.688 ± SD 0.030)



Correct on the gold standard. The rule is a lead, not proof: the deviation-to-delay link is weak and unstable.

Limitations and Future Work

Limitations

- Tree unstable across splits
- Root split hits 87% of traces: net too strict?
- Cutoff hides the size of the delay; one dataset; control flow only
- **Why weak:** no case is deviation-free, half of the deviation counts are pauses, resumes or aborts, and alignments cannot see waiting time.

Future work

- A Petri net that allows repeated pauses
- Time-aware features, e.g. days waited after a pause
- Regression tree on the continuous throughput time
- Data-aware checking (e.g. loan amount)

References: Dees, de Leoni, Mannhardt (2017). van der Aalst (2016) Process Mining. Berti et al. (2019) PM4Py. van Dongen (2017).